A Data Warehouse–Driven Analysis of Insurance Profitability, Risk, and Fraud at Analytic Mutual Insurance

BAN 621

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Team 8

Business Problem

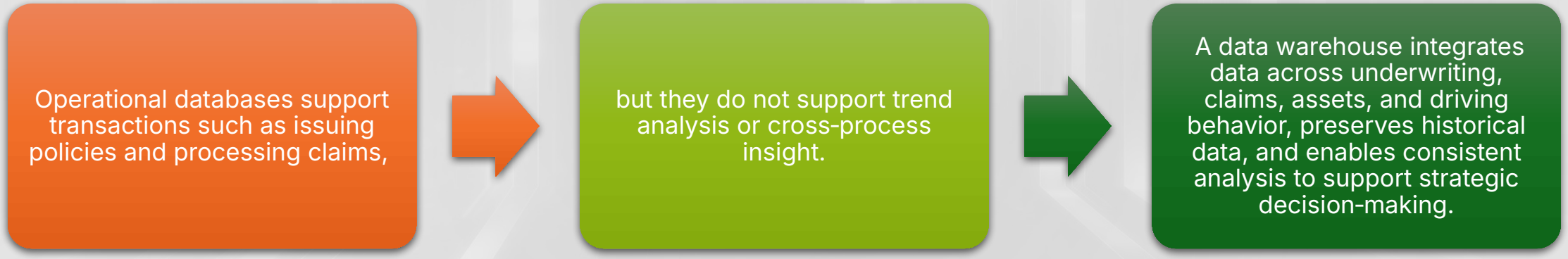
Analytic Mutual Insurance operates across homeowner and automobile insurance lines.

generating large volumes of policy, claims, and customer data.

However, this data exists in operational systems that are not designed for integrated, historical analysis.

As a result, management lacks visibility into profitability, risk exposure, claims behavior, and potential fraud across customers, regions, and policy types.

Why a Data Warehouse Is Needed



Data & Method



The data warehouse integrates policy, customer, claims, property, automobile, and driving behavior data.



For this project, realistic sample data was used to simulate operational data loading.



Data was modeled using dimensional modeling principles and implemented in SQL Server to support analytical querying.

Data Warehouse Design



The warehouse uses a **constellation schema** composed of multiple fact tables representing distinct business processes, including homeowner premiums, automobile premiums, claims, and property exposure.

Shared dimension tables such as customer, policy, date, and location provide consistent analytical context across all facts, enabling integrated analysis across lines of business.

ETL Process:

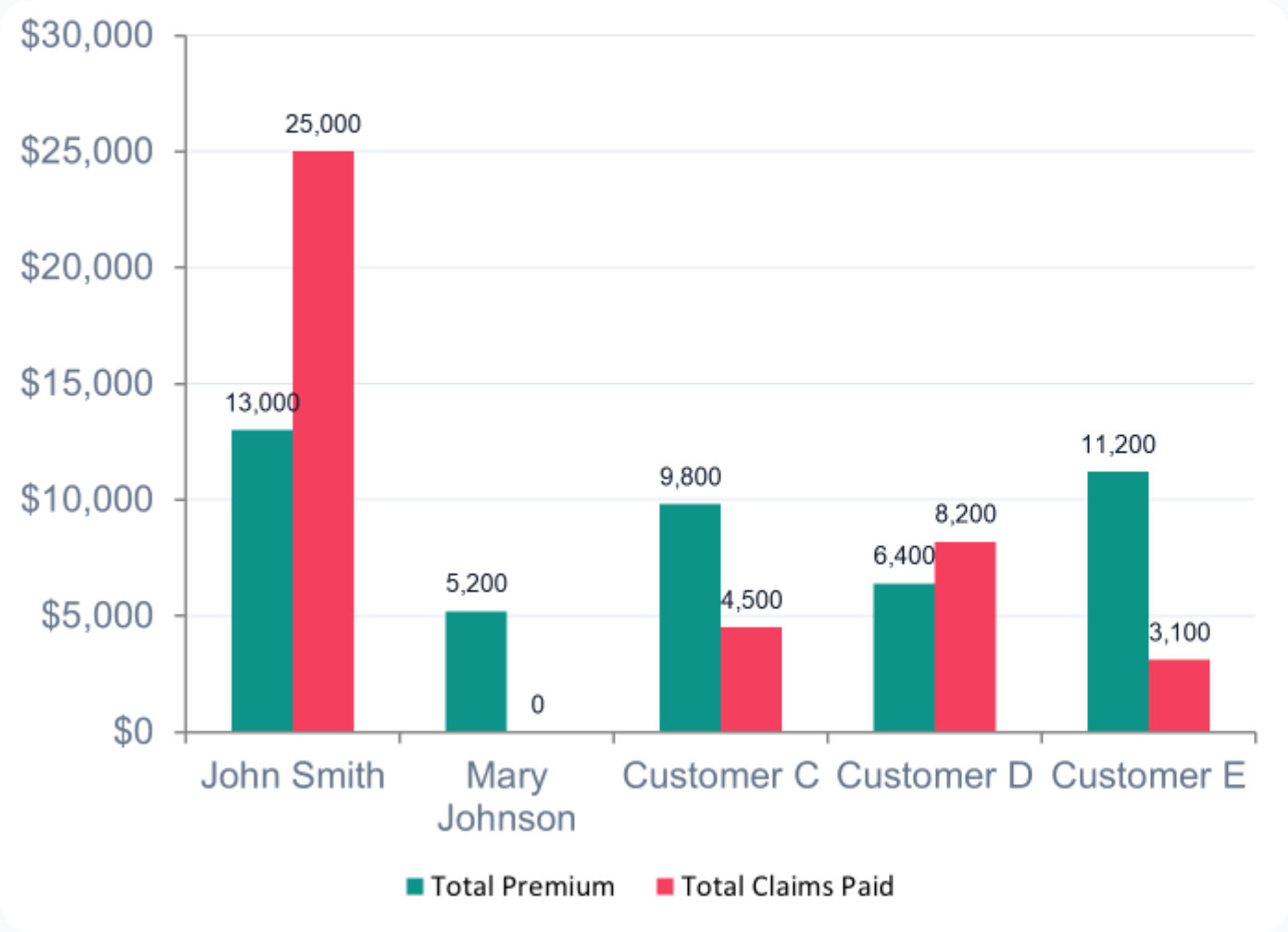
The ETL process was implemented using SQL insert statements to simulate data loading.

Dimension tables were populated first to establish descriptive context, followed by fact tables containing quantitative measures.

Referential integrity was enforced through primary and foreign key constraints to ensure data consistency and analytical accuracy.

Insight 1: Customer Profitability Analysis

Premium income vs. claims paid — underwriting P&L by customer



⚠ Underwriting Loss

John Smith's claim ($25K) exceeds his total premium ($13K) — a net underwriting loss of $12,000.

✓ Profitable Customer

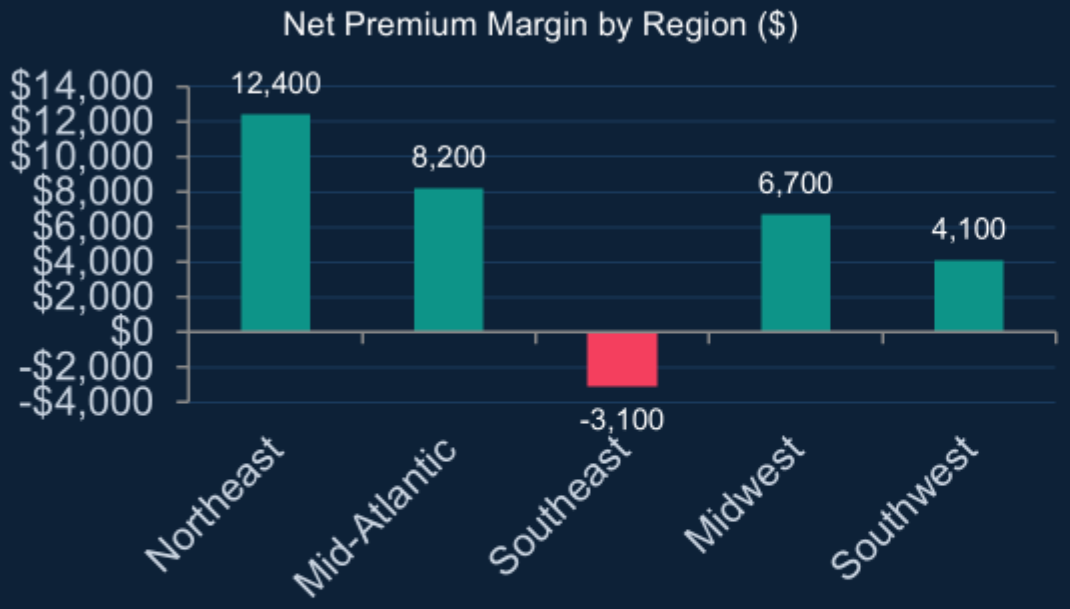
Mary Johnson: $5,200 premium, zero claims — 100% underwriting profit contribution.

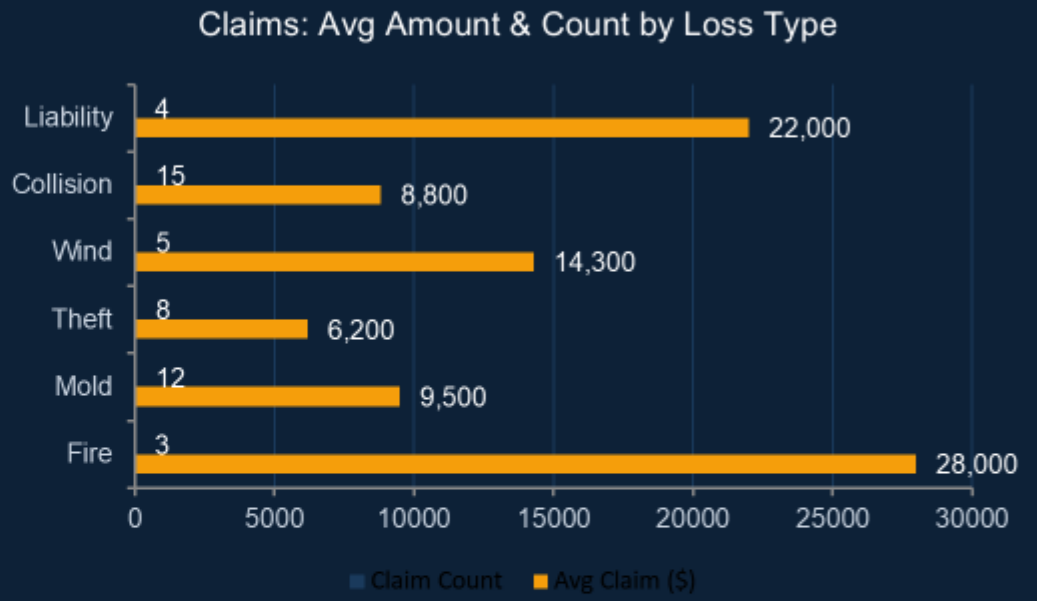
Key Action

Reprice or non-renew high-loss customers. Reward low-risk customers with loyalty discounts.

Insights 2 & 3: Regional Performance & Claims by Loss Type

Geographic underwriting margins and claims frequency/severity





Southeast Deficit

Negative net margin — high claims or low premium rate adequacy. Immediate underwriting review recommended.

Fire: High Severity

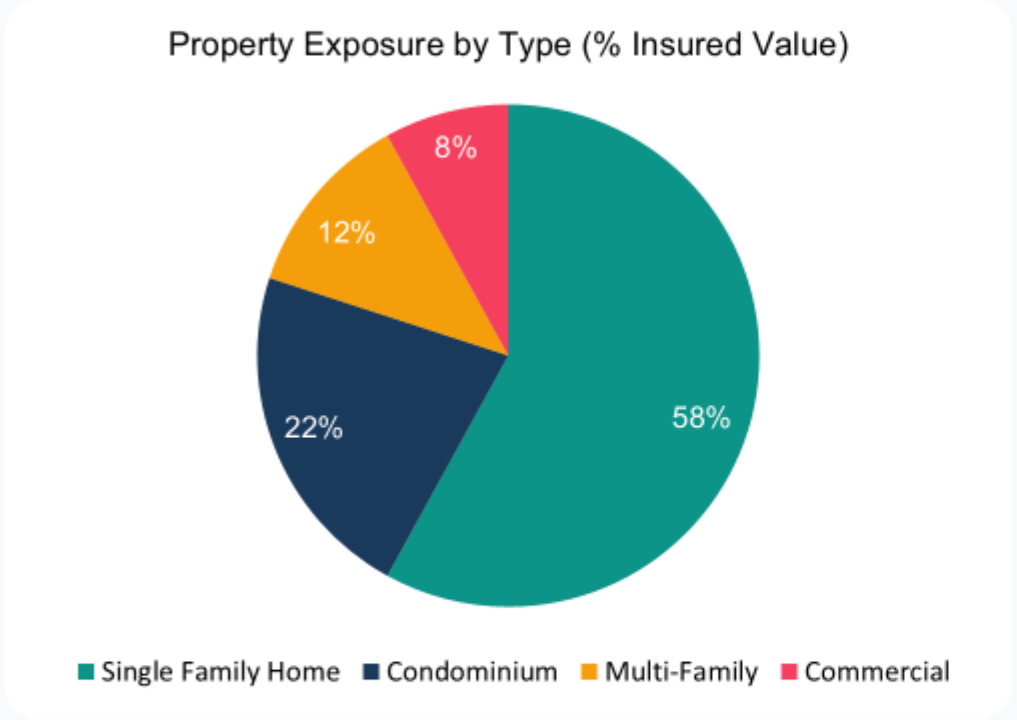
Highest average claim ($28K) despite low frequency (3 claims). Reserve adequacy must reflect catastrophic tail risk.

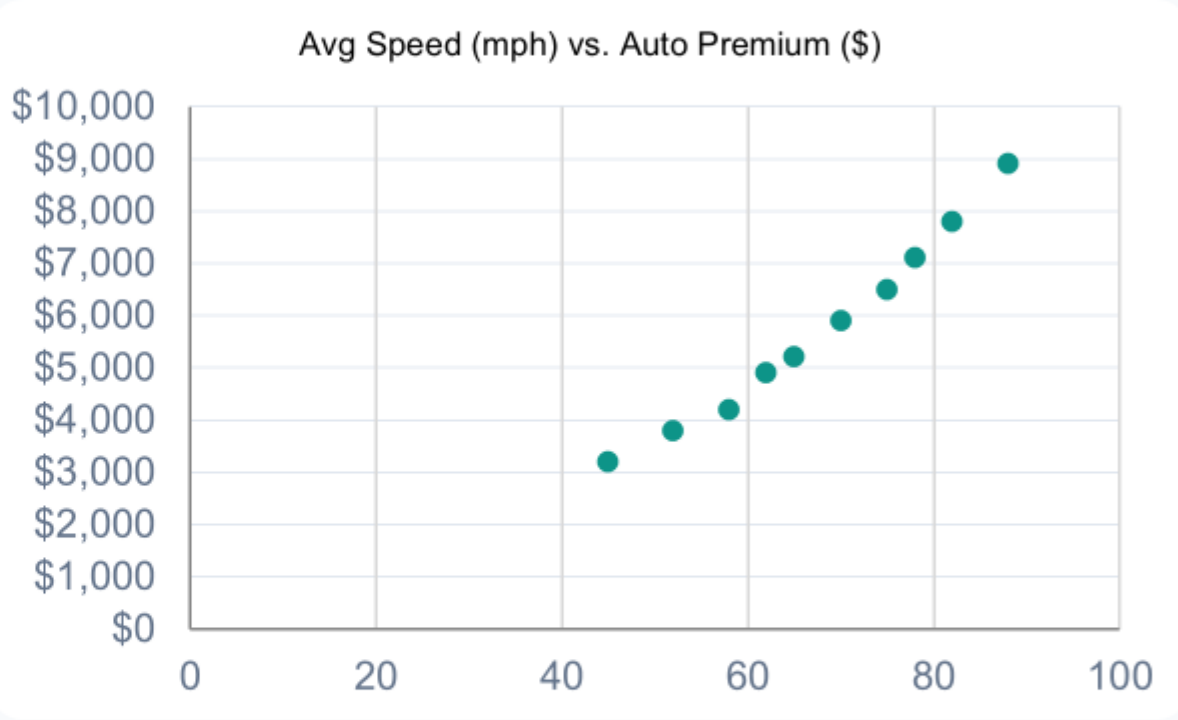
Collision: High Freq

15 collision claims — highest frequency loss type. Telematics pricing adjustment can improve margin here.

Fire and Mold sit in the dangerous "low
Insights 4 & 5: Property Exposure & Driving Behavior

Risk concentration and telematics-driven premium analysis





58%
Single Family
Dominates Exposure

65.5 mph
Avg Speed on
Top Premium Policy

42.3 mi
Avg Miles/Day
Highest Premium Driver

18/day
Avg Braking Events
High-Risk Auto

Insight 6: Fraudulent Claim Identification

SQL-driven fraud flag logic — zero reserve & zero payment

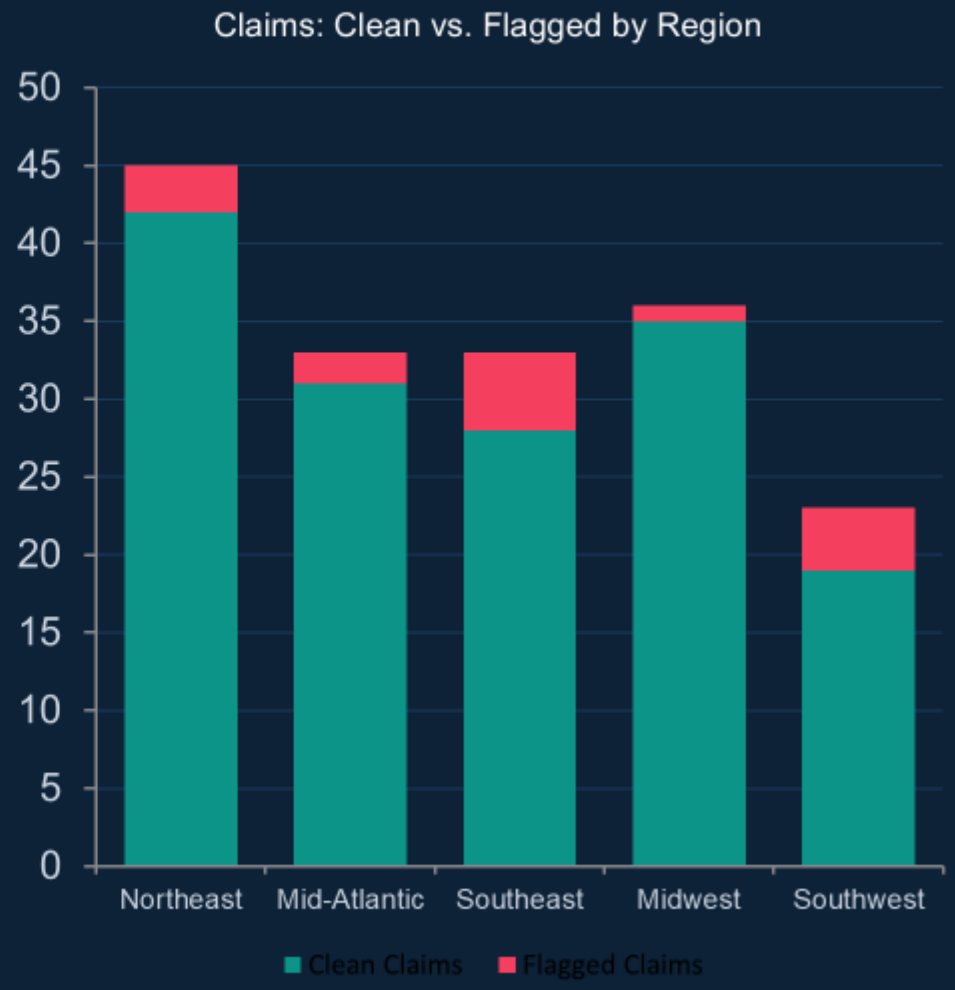
Fraud Flag Criteria

A claim is flagged as potentially fraudulent when:
· ReserveAmount = $0 AND
· ClaimAmountPaid = $0
This matches the company's internal definition of claims that were filed but never processed — a key fraud indicator.

Undetected fraudulent claims inflate reported claim counts, distort loss ratios, and erode underwriting profit. Early identification enables SIU (Special Investigations Unit) referral before reserves are set.

Warehouse Advantage

Because Fact_Claim retains both ReserveAmount and ClaimAmountPaid at the claim grain, the fraud detection view can be run in real-time — impossible with fragmented legacy systems.



Business Value:



IMPROVE UNDERWRITING AND PRICING DECISIONS



MANAGE EXPOSURE MORE EFFECTIVELY



STRENGTHEN CLAIMS OVERSIGHT.



SUPPORT PROFITABILITY



RISK MANAGEMENT



AND OPERATIONAL EFFICIENCY.

Conclusion & Next Steps:



Refine underwriting criteria based on profitability and regional risk, monitor property exposure concentrations, incorporate driving behavior into pricing models



Expand fraud detection analytics.



Future enhancements could include real-time data feeds, predictive analytics, and executive dashboards to further strengthen decision-making